


CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

Test Pattern

JEE(Advanced)
FULL SYLLABUS
10-04-2024
JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

Time : 3 Hours

PAPER-2 (OPTIONAL)

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY
GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 18)

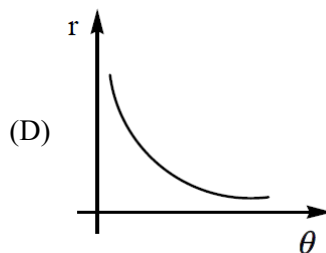
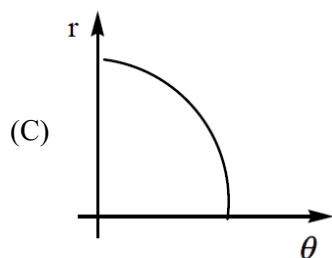
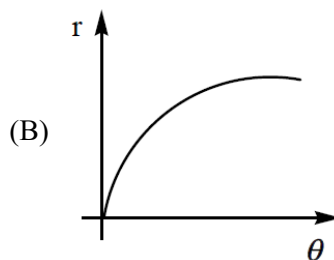
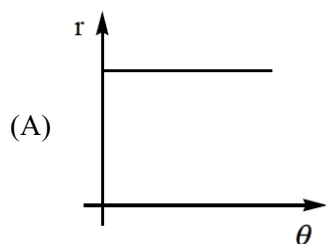
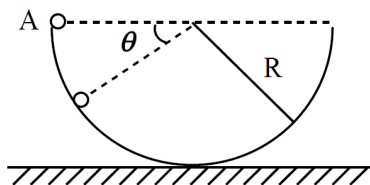
- This section contains **SIX (06)** questions.
- Each question has Four options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. A small particle of mass m is released from rest from point A inside a fixed smooth hemispherical bowl as shown in the figure. The ratio (r) of magnitude of centripetal force and normal reaction on the particle at any point varies with θ as



2. A simple pendulum consists of a bob of mass m and thread of length ℓ . In an experimental demonstration, your physics teacher pulls the bob aside until the thread makes an angle $\theta_0 = \left(\frac{2}{\sqrt{3}} - 1 \right)$ radians with the vertical. Now he pushes the bob towards the equilibrium position. While pushing, he continuously applies a force $F = mg$ along the circular arc until the thread comes in vertical position and thereafter he releases the bob. For what approximate length of time t_0 did he push the bob ?

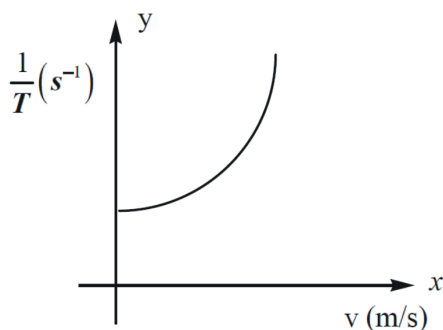
(A) $\frac{\pi}{6} \sqrt{\frac{\ell}{g}}$

(B) $\pi \sqrt{\frac{\ell}{g}}$

(C) $\frac{\pi}{3} \sqrt{\frac{\ell}{g}}$

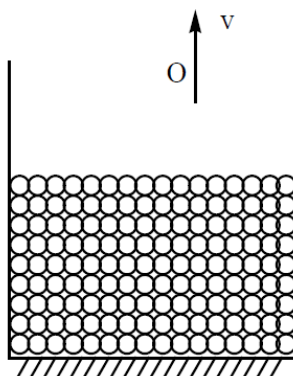
(D) $\frac{2\pi}{3} \sqrt{\frac{\ell}{g}}$

3. An engineer designs a robot that can climb stairs. Time T in which battery used in the robot is fully discharged depends on its constant speed v relative to the stairs. This dependence is shown in the graph which follows $y - 0.04 = x^2$. Determine the maximum length of the staircase that the robot can climb with constant velocity.

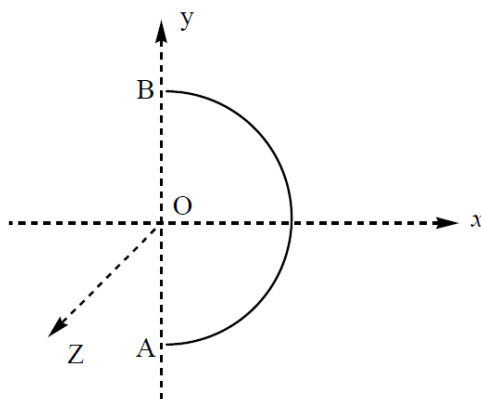


- (A) 2.5m
(B) 5m
(C) 7.5m
(D) 1.25m

4. From a box of balls stacked as shown, balls are removed at equal intervals of time with a speed v as shown. Average velocity of centre of mass of remaining balls in the box is



- (A) zero
(B) upward
(C) downward
(D) cannot be determined
5. There are 2016 points in space. Each pair of points is connected by a resistance of 1Ω . The equivalent resistance between any two points is
- (A) $\frac{1}{2016}\Omega$
(B) $\frac{2}{2016}\Omega$
(C) $\frac{1}{2015}\Omega$
(D) $\frac{2}{2015}\Omega$
6. A part of a conductor carrying current I is in the form of a semicircle AB of radius R and lying in xy -plane with its centre at origin as shown in figure. Which of the following is the magnitude of $\oint \vec{B} \cdot d\vec{l}$ for the circular loop along $3x^2 + 3z^2 = R^2$ in xz -plane due to current in curve AB ?



- (A) $\frac{\sqrt{3}}{2}\mu_0 I$
(B) $(\sqrt{3} + 1)\frac{\mu_0 I}{2}$
(C) $(2 - \sqrt{3})\frac{\mu_0 I}{2}$
(D) Zero

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

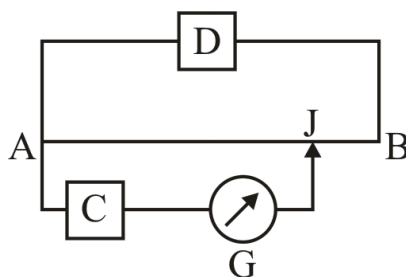
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

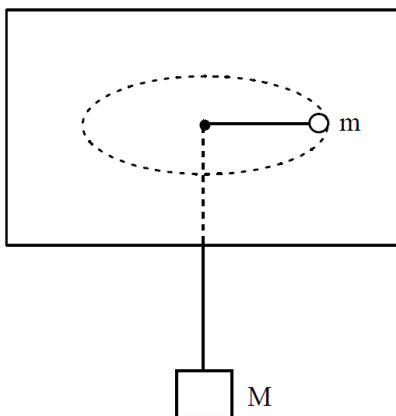
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. The figure shows a potentiometer arrangement. D is the driving cell and C is the cell whose emf is to be determined. AB is the potentiometer wire and G is galvanometer. J is sliding contact which can touch any point on AB. Which of the following are essential conditions for obtaining balance ?

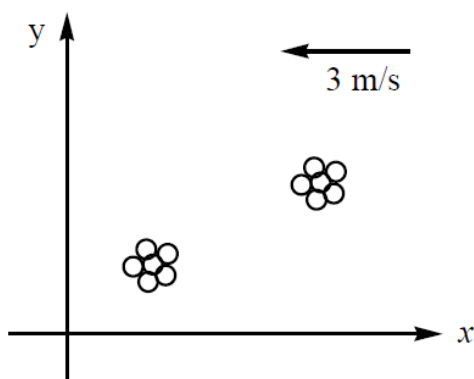


- (A) The emf of D must be greater than the emf of C
- (B) Either the positive terminals of both D and C or the negative terminals of both D and C must be joined to A
- (C) The positive terminals of D and C must be joined
- (D) The resistance of G must be less than resistance of AB

8. A small disc of mass m is attached to the end of a light inextensible cord, which passes through a hole in a frictionless horizontal tabletop. At the other end of the cord, a weight of mass M is attached. Initially disc is moving on a circle of radius R with angular velocity ω . If the hanging weight is pulled slightly down by x_0 ($\ll R$) and then released,

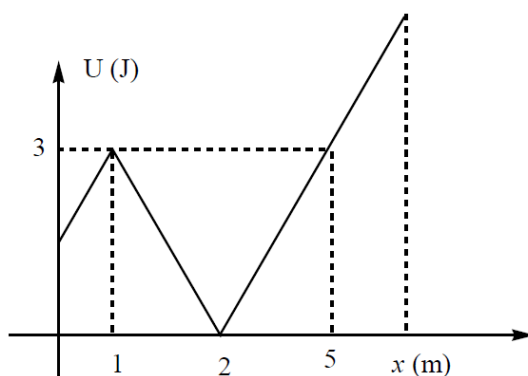


- (A) The net force on m at an instant when M moves up by $\frac{x_0}{2}$ after the release is $Mg + \frac{3x_0mM\omega^2}{2(M+m)}$
- (B) The net force on M at an instant when M moves up by $\frac{x_0}{2}$ after the release is $Mg + \frac{3x_0mM\omega^2}{2(M+m)}$
- (C) The net force on M at an instant when M moves up by $\frac{x_0}{2}$ after the release is $\frac{3x_0mM\omega^2}{2(M+m)}$
- (D) Acceleration of m at an instant when M moves up by $\frac{x_0}{2}$ after the release is $\frac{3x_0m\omega^2}{2(M+m)}$
9. Avantika fought her kidnappers with all her might, but to no avail. In her tussle, two flowers (at two different instants) from her bracelet fell in to the river as the boat fled with a constant velocity towards an unknown destination. Probably, Admiral Kattappa was too late, or was he? All he could see in the river were two flowers that once decorated the delicate wrist. The co-ordinates of the flowers were $(3m, 3m)$ and $(7m, 6m)$. Given the river flows from right to left at $3m/s$ and the speed of the boat with respect to water was $5m/s$, in which possible direction(s) w.r.t a fixed xy -axis shown in figure did the kidnappers flee? (Neglect dimensions of the boat)

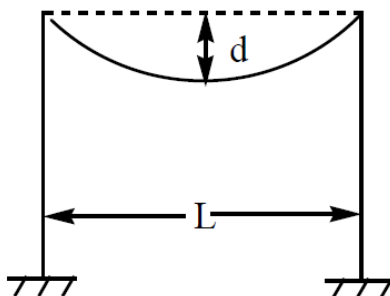


- (A) $\hat{i} + 3\hat{j}$ (B) $-3\hat{i} + \hat{j}$
- (C) $-7\hat{i} - 3\hat{j}$ (D) $\hat{i} - 3\hat{j}$

10. A particle of mass $m = 2\text{kg}$ with potential energy shown in the graph is moving towards positive x -axis with a speed 1m/s at $x = 1\text{m}$.

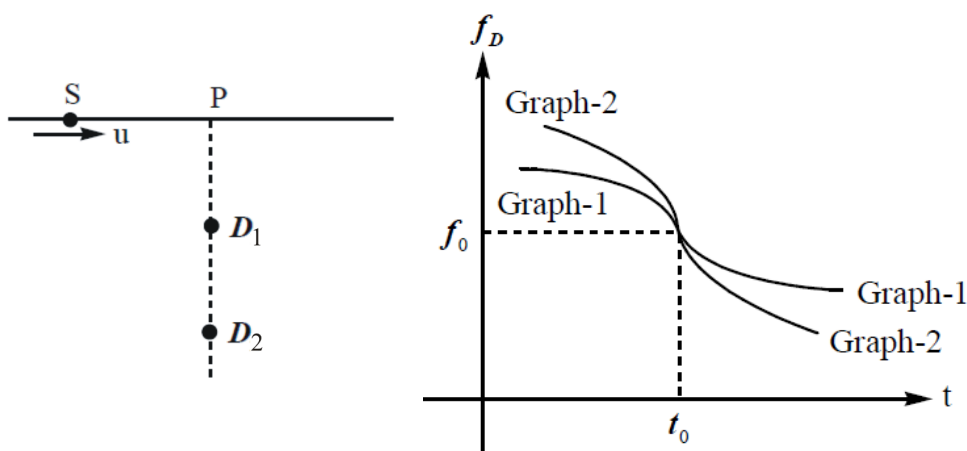


- (A) Particle reverses its direction of motion at $x = 2\text{m}$
- (B) Particle reverses its direction of motion at $x = 6\text{m}$
- (C) Force acting on the particle at the instant of its direction of reversal is 1N towards negative x direction
- (D) Maximum speed of the particle before it reverses its direction is 3m/s
11. The cable strung between two neighbouring electricity posts sags a little. The mass per unit length of the cable is λ , the distance between the posts is L , and the 'maximum sag' of the cable is d ($d \ll L$). Due to relatively small sag the tension in the cable can be assumed to be uniform. The cable obeys Hooke's law. Choose the correct statement(s) from the following:



- (A) The approximate value of tension is $\frac{\lambda L^2 g}{4d}$
- (B) The approximate value of tension is $\frac{\lambda L^2 g}{8d}$
- (C) The appropriate formula for calculation of elastic strain energy stored in the cable is $\frac{1}{2} \times \text{stress} \times \text{strain} \times \text{volume}$
- (D) The appropriate formula for calculation of elastic strain energy stored in the cable is $\text{stress} \times \text{strain} \times \text{volume}$

12. The diameter of a cylinder is measured using a vernier calipers with no zero error. While measuring diameter of a cylinder, it is found that the zero of vernier scale lies between 5.10 cm and 5.15 cm of the main scale. Vernier scale has 50 divisions whose total length is equivalent to 2.45 cm. The closest division of the vernier scale that almost coincides with a main scale division is 24th VSD but this 24th VSD actually falls slightly to the right of the main scale division. The diameter of cylinder may be (each MSD = 0.05cm)
- (A) 51.238 mm (B) 51.236 mm
(C) 51.242 mm (D) 51.246 mm
13. A submarine travels horizontally with a uniform speed 'u' inside a calm sea. Then the flow around the submarine is
- (A) unsteady as seen by a stationary fish in the sea
(B) steady as seen by a crab moving with same velocity as that of the marine
(C) unsteady as seen by the captain inside the marine
(D) steady as seen by a person moving down in the sea
14. A source 'S' emitting sound of frequency f moves uniformly along a straight line with a velocity u as shown. The velocity of sound in the medium is $v(>u)$. Two detectors D_1 and D_2 are placed on the line perpendicular to line of motion of S. Point P represents the point of closest approach of S to the detectors. The frequency detected (f_D) by D_1 and D_2 versus time t graph is shown on the same graph paper.



Choose the incorrect option(s)

- (A) Graph-1 belongs to detector D_1
(B) $t = t_0$ is the instant when source S is at P
(C) The value of f_0 is f
(D) The value of f_0 is $\left(\frac{v^2}{v^2 - u^2} \right) f$

SECTION-I (iii) : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

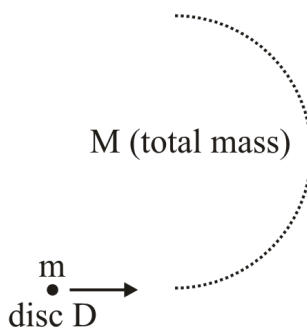
Full Marks : +4 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

Paragraph for Questions 15 and 16

On an air-hockey table, there are N identical small discs lying equally spaced around a semicircle; the total mass of the discs is M . Another small disc, D , of mass m , travelling in a direction perpendicular to the closing diameter of the semicircle, strikes the first of the stationary discs. By some miracle, it subsequently bounces off all of the other $N - 1$ discs in turn, after which it is travelling in a direction directly opposed to that of its initial motion. All the collisions are perfectly elastic, and friction is everywhere negligible.



15. In the limiting case of $N \rightarrow \infty$, what is the minimal value of the mass ratio M/m for such a miracle to be possible?
- (A) π (B) $e^{\pi/2}$
 (C) $4 \ln 2$ (D) 2
16. When the mass ratio has the critical value found in above question what is the ratio of the final and initial speeds of D ?
- (A) $\ln 2$ (B) $e^{-\pi}$
 (C) 1 (D) $\frac{1}{2}$

Paragraph for Questions 17 and 18

Consider first a metal sphere of radius R and total charge Q , which (because of symmetry) is distributed uniformly on its surface, i.e. the surface charge density is $\sigma_0 = \frac{Q}{4\pi R^2} = \text{constant}$.

The total field effects of the charges on the small surface pieces opposite each other cancel out in pairs, so the net electric field of all the charges on the sphere is zero at any point P inside the sphere. The same is shown in figure-1.

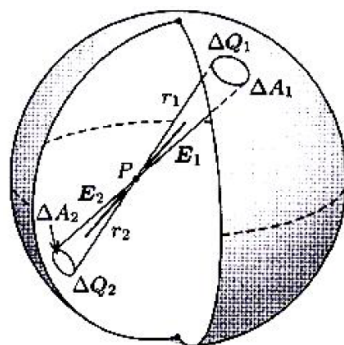


Fig. 1

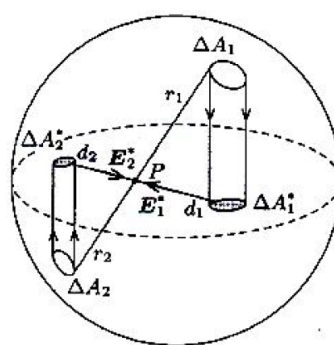


Fig. 2

Now let us move on to the case of the electrically charged metal disc. We show that its charge distribution can be found from the uniform surface density of the charged metal sphere. Let us (notionally) 'glue into position' the charges on the surface of the sphere, and then project the points of the spherical surface perpendicularly onto a plane that contains the centre of the sphere and point P investigated earlier.

As the charges cannot move, the charge ΔQ_1 , originally on the surface area ΔA_1 , when projected onto the plane, occupies a somewhat smaller area ΔA_1^* , and so the surface charge density (with an unchanged amount of charge) increases. In addition, its distance from the point P also changes, from the original value of r_1 to the smaller one of d_1 . The corresponding thing happens with the charge ΔQ_2 originally located on the opposite side of the sphere.

17. Consider the following statements about figure-1 and figure-2

(i) The electric field produced at P by the two charged pieces (projected onto the plane and still glued to the disc) in the figure-2 is zero.

(ii) $\frac{\Delta A_1}{\Delta A_2} = \frac{r_1}{r_2}$

(iii) $\frac{\Delta A_1}{\Delta A_2} = \frac{d_1^2}{d_2^2}$

(iv) If a metal disc is charged by Q , the charge distribution on it is the same way as it would be if that on a uniformly charged spherical surface were projected perpendicular on it as shown in figure-2.

The correct statement(s) is/are

- (A) (i), (ii), (iii) (B) (i), (iii), (iv) (C) (i), (ii), (iv) (D) (i), (ii), (iii), (iv)

18. A large thin metal disc of radius R is charged and carries a total electric charge of Q . The charge density σ on the disc at distance r from centre far away from the rim of the disc is (Neglect edge effects)

(A) $\frac{Q}{2\pi R^2} \frac{R}{\sqrt{R^2 - r^2}}$

(B) $\frac{Q}{4\pi R^2} \frac{R}{\sqrt{R^2 + r^2}}$

(C) $\frac{Q}{4\pi R^2} \frac{R}{\sqrt{R^2 - r^2}}$

(D) $\frac{Q}{2\pi R^2} \frac{R}{\sqrt{R^2 + r^2}}$

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 18)

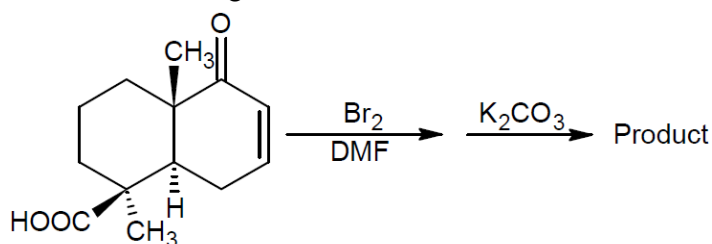
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Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

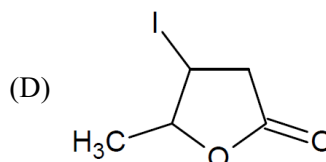
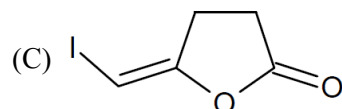
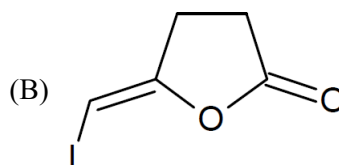
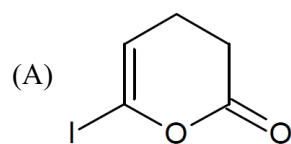
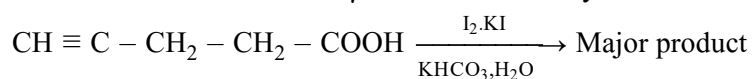
Negative Marks : -1 In all other cases

1. Identify the major product in the following reaction?

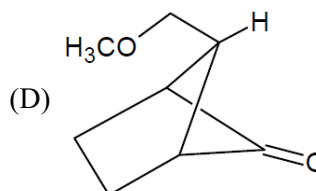
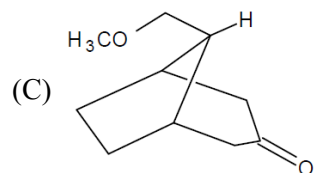
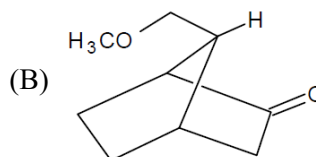
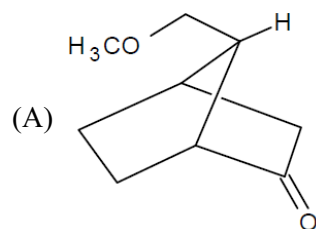
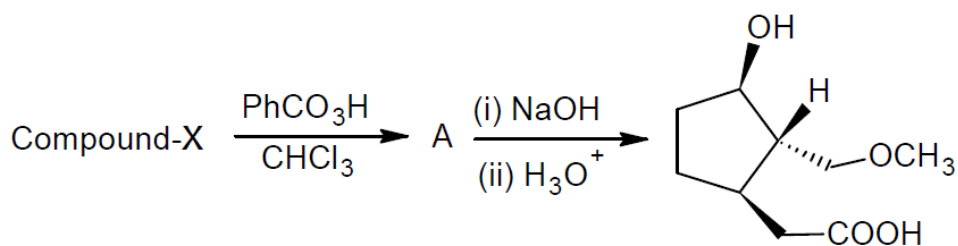


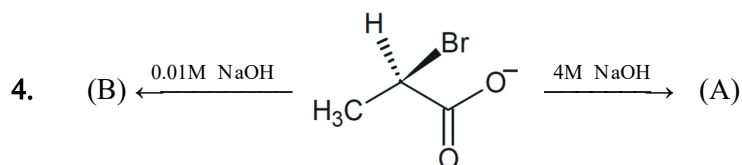
- (A)
- (B)
- (C)
- (D)

2. What is the structure of the γ -lactone formed by iodolactonisation of 4-pentynoic acid?



3. A key step in the synthesis of an important class of lipids prostaglandins involves the following sequence of reactions. Identify the reactant 'x'?





Which of the following is correct regarding compounds (A) and (B)

- (A) (A) and (B) are super imposable mirror images
- (B) The configuration of (A) is 'R' and (B) is 'S'
- (C) (A) and (B) are diastereomers
- (D) (A) is formed with inversion of configuration & (B) with retention of configuration
5. Basing on stability of the lower species to which they are reduced, identify the correct order of oxidising power.
- (A) $\text{MnO}_4^- < \text{VO}_2^+ < \text{Cr}_2\text{O}_7^{2-}$
- (B) $\text{Cr}_2\text{O}_7^{2-} < \text{VO}_2^+ < \text{MnO}_4^-$
- (C) $\text{VO}_2^+ < \text{MnO}_4^- < \text{Cr}_2\text{O}_7^{2-}$
- (D) $\text{VO}_2^+ < \text{Cr}_2\text{O}_7^{2-} < \text{MnO}_4^-$
6. Quartz + Carbon $\xrightarrow[\Delta]{\text{Cl}_2}$ $\text{A}_{(\ell)} + \text{B}_{(\text{g})}$
- $\text{A} + \text{CH}_3\text{MgBr} \xrightarrow{\text{H}_3\text{O}^+} \text{C}$
- $\text{C} \xrightarrow{\Delta} \text{D}$. Identify the correct statement(s) regarding the reaction sequence.
- (A) B is formed when acetic acid is heated with conc. H_2SO_4
- (B) D is a chain silicone
- (C) D can also be prepared by heating orthosilicic acid.
- (D) D is a cross linked silicone polymer

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

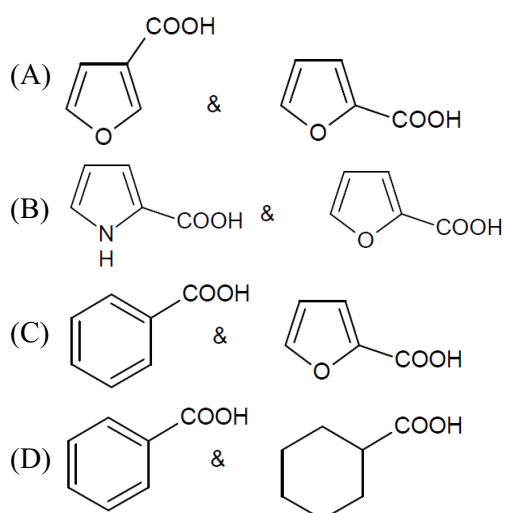
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

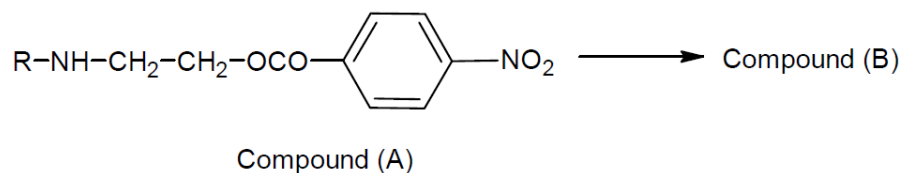
Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

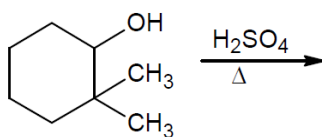
7. Identify the pair of acids in which later is more acidic than former?

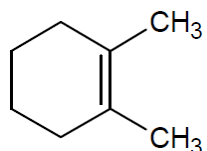
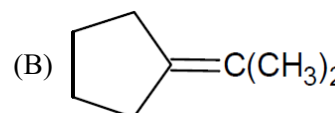
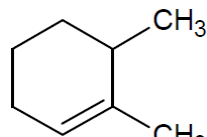
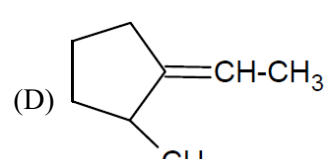


8. When the compounds of the following type are allowed to stand in pentane, they are converted into a constitutional isomer. Which of the following is/are correct about the constitutional isomers (A) and (B)?

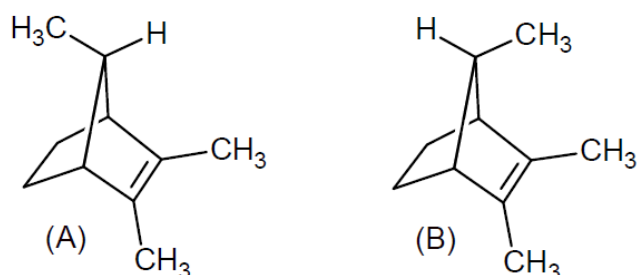


- (A) (A) & (B) are functional isomers
 (B) Both give same products on hydrolysis
 (C) Hydrolysis products can be separated by dil.HCl
 (D) Hydrolysis products can be separated by dil.NaOH
9. What is/are the possible product/s in the following dehydration reaction?



- (A)  (B) 
 (C)  (D) 

10. Which of the following statement/s is/are correct about compounds (A) and (B)?



- (A) The rate of hydrogenation of (A) is more than (B) under similar conditions
 (B) The compounds (A) and (B) are stereo isomers.
 (C) Catalytic hydrogenation of (A) or (B) gives two diastereomers in unequal ratio.
 (D) The pair of diastereomers can be resolved in to optically active compounds.

11. Among the following the correct statement(s) is(are)
- (A) The principle involved in 'zone refining' is fractional distillation
 - (B) Parke's process in the metallurgy of silver is an application of Distribution Law
 - (C) Preferential adsorption is the principle of froth-floatation process.
 - (D) Open-Hearth furnace works on regenerative principle of heat economy.
12. Identify the incorrect statement(s)
- (A) In acidic medium, both manganate and chromate ions undergo disproportionation.
 - (B) Manganate ion is paramagnetic while permanganate ion is diamagnetic
 - (C) Out of trivalent ions of Mn, Cr, V and Ti, the relatively more stable one is Mn^{3+} ion
 - (D) Higher oxides like Mn_2O_7 , CrO_3 and V_2O_5 are predominantly basic in nature.
13. Which of the following statements is(are) correct ?
- (A) $\text{XeO}_3 + 2\text{XeF}_6 \rightarrow 3 \text{ 'x' }$, the structure of x is square pyramidal
 - (B) $\text{XeO}_3 + \text{XeOF}_4 \rightarrow 2 \text{ 'y' }$, the structure of y is see-saw
 - (C) XeF_2 , XeF_4 , and XeF_6 react with H_2 to give Xe
 - (D) XeF_6 does not react with SiO_2
14. Under certain conditions of concentration and temperature HNO_3 reacts with Zn and its reduction products are NO_2 and NO in a molar ratio 1 : 3. How many moles of HNO_3 are consumed by 1 mol of Zn?
- (A) 2.2
 - (B) 2.4
 - (C) 2.8
 - (D) 3.0

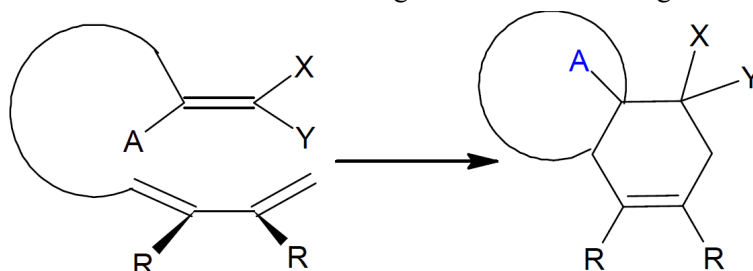
SECTION-I (iii) : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct option is chosen.
 Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
 Negative Marks : -2 In all other cases.

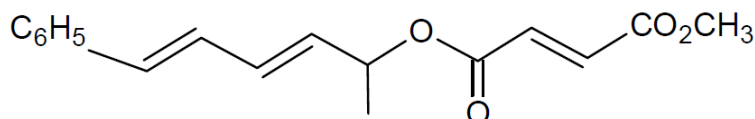
Paragraph for Questions 15 and 16

The Diels-Alder reaction, a concerted [4+2]-cycloaddition between a diene and an olefin to yield a cyclohexene, was discovered in 1928 by Prof. Otto Diels and his coworker Kurt Alder. In Diels-Alder reaction the diene and the dienophile are functional groups in separate molecules. The Diels-Alder reaction is a concerted, one-step reaction that proceeds with high stereospecificity. The reaction can be intermolecular or intramolecular. An intramolecular generates two new rings in a single operation.



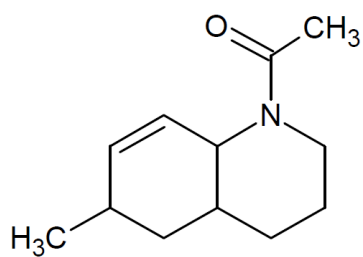
Diels-Alder reactions are reversible. Cyclohexenes can dissociate into a diene and an alkene by retro Diels-Alder reaction.

15. What is the structure of the intramolecular Diels-Alder product of the following compound?



- (A)
- (B)
- (C)
- (D)

16. What compound would you use to prepare the following compound by an intramolecular Diels-Alder reaction?



- (A)
- (B)
- (C)
- (D)

Paragraph for Questions 17 and 18

Solvents have been classified in a number of ways depending on their physical and chemical properties. An inorganic nonaqueous solvent is a solvent other than water. Common examples are liquid ammonia, liquid sulphurdioxide, sulphuryl chloride, dinitrogen tetroxide antimony trichloride, bromine trifluoride, hydrogen fluoride etc. These are found to be electrically conducting and therefore expected to undergo self – ionisation. These solvents are used in chemical research and industry for reactions that cannot occur in aqueous solutions.

17. Pick out correct statements among the following
- (I) Dissolution of BrSbF_8 or KBrF_4 in liquid BrF_3 gives a highly conducting solution.
 - (II) BrSbF_8 acts as an acid in BrF_3
 - (III) KBrF_4 acts as a base in BrF_3
 - (IV) The reaction, $\text{AgBrF}_4 + \text{BrF}_2\text{SbF}_6 \rightarrow \text{AgSbF}_6 + 2\text{BrF}_3$, is a neutralization reaction in BrF_3
- (A) Only I and II
(B) Only I and III
(C) Only I, II and III
(D) All are correct
18. Which of the following statement is correct ?
- (A) BrF_5 is strong fluorinating agent than BrF_3
 - (B) KF acts as acid and SbF_5 acts as base in liquid BrF_3 .
 - (C) The liquid I_2Cl_6 has appreciable conductivity due to self ionisation into I^+ , ICl_6^-
 - (D) BrF_5 under goes disproportionation when dissolved in water

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has Four options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. If z is a non real complex number satisfying $z^{26} = 1$, then $\sum_{r=1}^{25} \frac{1}{1 + z^{9r} + z^{18r}} =$
 - (A) 14
 - (B) 15
 - (C) 16
 - (D) 17
2. Let the normal at $(2 \cos \alpha, \sin \alpha)$ to the ellipse $x^2 + 4y^2 = 4$ intersects the ellipse again at $(2 \cos \beta, \sin \beta)$ ($\alpha, \beta \in \mathbb{R}$). If $\sin \beta = -\sin \alpha \left(\frac{a + b \sin^2 \alpha}{c + d \sin^2 \alpha} \right)$ in simplest form, then the value of $(a+b-c-d)$ is equal to ($a, b, c, d \in \mathbb{I}$)
 - (A) -3
 - (B) -2
 - (C) 0
 - (D) 4
3. If $I = \int_0^{4\pi} \log_e |13 \sin x + 3\sqrt{3} \cos x| dx$, then the value of I is equal to
 - (A) $\log_e^7$
 - (B) $2\pi \log_e^7$
 - (C) $4\pi \log_e^7$
 - (D) $8\pi \log_e^7$

4. Let $P_1(x) = \alpha x^2 - \beta x - \gamma$, $P_2(x) = \beta x^2 - \gamma x - \alpha$, $P_3(x) = \gamma x^2 - \alpha x - \beta$ be three quadratic polynomials where α, β, γ are non-zero real numbers suppose there exists a real number k such that $P_1(k) = P_2(k) = P_3(k)$ then $\frac{9\alpha + 8\beta + 5\gamma}{5\alpha - 3\beta - \gamma} = \underline{\hspace{2cm}}$
- (A) 19
(B) 15
(C) 21
(D) 22
5. Area enclosed by $y = g(x)$, $x = -3$, $x = 5$ and x -axis where $g(x)$ is the inverse function of $f(x) = x^3 + 3x + 1$ is _____
- (A) $\frac{5}{2}$
(B) $\frac{7}{2}$
(C) $\frac{9}{2}$
(D) $\frac{3}{2}$
6. If (α_r, β_r) where $r = 1, 2, 3, \dots$ are the integral solutions of the equation $2x^2 y^2 + y^2 - 6x^2 - 12 = 0$, then the area of the polygon whose vertices are (α_r, β_r) , $r = 1, 2, 3, \dots$ is
- (A) 12 sq. units
(B) 16 sq. units
(C) 20 sq. units
(D) 18 sq. units

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
7. If a, b, c are positive real numbers such that $a + 2b + 3c = 15$ and the maximum value of $6(1+a)bc + a(2b+3c)$ is k , then which of the following is/are true ?
- (A) The sum of the digits of k is 2
- (B) The total number of positive integral divisors of k is 12
- (C) The number formed by last two digits of k is divisible by 23
- (D) The total number of odd positive integral divisors of k is 3

8. If $I_n = \int_0^{\pi/2} x(\sin x + \cos x)^n dx$, and $\frac{nI_n - \frac{\pi}{2}}{I_{n-2}} = f(n)$, then which of the following is/are true ?
- (A) The sum of the digits of $f(2016)$ is 7
- (B) $f(2017)$ is divisible by 63
- (C) $f(2017)$ is divisible by 32
- (D) $f(2016)$ is divisible by 31
9. If $f : \left(0, \frac{\pi}{2}\right) \rightarrow \mathbb{R}^+$ be a real valued function satisfies the differential equation $xf(x) = 1 + f(x)(x^2f(x) - 1)$ and $f\left(\frac{\pi}{4}\right) = \frac{4}{\pi}$ and also $A = \int_0^{\frac{\pi}{2}} (\cos x) f(x) dx$ then which of the following is/are true ?
- (A) $\frac{2}{\pi} < A < 1$
- (B) $1 < A < \frac{\pi}{2}$
- (C) $\frac{7}{6} < A < \frac{\pi+6}{6}$
- (D) $\frac{1+\sqrt{2}}{2} < A < \frac{\pi+2\sqrt{2}}{4}$
10. If $P = \lim_{n \rightarrow \infty} \prod_{r=1}^n \left(1 + \frac{1}{4r^2 - 1}\right)$, then $[37P]$ is greater than or equal to, ($[.]$ denotes greatest integer function)
- (A) 55
- (B) 57
- (C) 58
- (D) 59

11. If P is a matrix of order 3×2 and Q is a matrix of order 2×3 such that $PQ = \begin{bmatrix} 8 & 2 & -2 \\ 2 & 5 & 4 \\ -2 & 4 & 5 \end{bmatrix}$ and also QP is an invertible matrix, then $|(QP)| \geq$ _____
- (A) 78
(B) 88
(C) 80
(D) 90
12. Let $f(x) = x^3 + 5x + 8$ and $x = \alpha$ be a point such that $f'(\alpha) \neq \frac{f(b) - f(a)}{b - a}$ for any two values of $a, b \in \mathbb{R}$, then the number of such points is n , where n is less than
- (A) 1
(B) 2
(C) 3
(D) 4
13. If ${}^{29}C_1 + 5 \cdot {}^{29}C_5 + 9 \cdot {}^{29}C_9 + \dots + 29 \cdot {}^{29}C_{29} = k$, then k is divisible by
- (A) 27
(B) 23
(C) 32
(D) 64
14. If PA and PB are two tangents drawn from P to the parabola $4y^2 = x$ and the normals drawn at A and B on the parabola are again intersect at a point Q on the same parabola. Then the locus of the centre of the circle passing through P, A and B is
- (A) An Ellipse whose eccentricity is $\frac{3}{5}$
(B) A parabola whose length of latusrectums is $\frac{1}{32}$
(C) An ellipse whose centre is $\left(\frac{1}{2}, \frac{3}{2}\right)$
(D) A Parabola whose vertex is $\left(\frac{1}{4}, 0\right)$

SECTION-I (iii) : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
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Negative Marks : -2 In all other cases.

Paragraph for Questions 15 and 16

Let $f(x) = ||x| - 1| - 1$ and $g(x) = \{\max(f(t) ; x \leq t \leq x + 1, x \in \mathbb{R})\}$ then answer the following

15. The number of points of non-differentiability of $g(x)$ is ____

- (A) 7 (B) 8
(C) 9 (D) 5

16. The graph of $g(x)$ is symmetric about the line

- (A) $x = \frac{1}{2}$ (B) $x = -\frac{1}{2}$
(C) $x = 0$ (D) $x = -1$

Paragraph for Questions 17 and 18

If $a, b, c > 0$ are such that $a + b + c = 3$ and two positive numbers $m, n \neq 1$ are such that $\max(a^m + b^m + c^m) = 3$ and $\min(a^n + b^n + c^n) = 3$ and a function $f(x) = x^2 + px + q$ is such that $f(x) = 0$ has m, n as its roots, then answer the following

17. Which is correct ?

- (A) $(p + q + 1)p < 0$ (B) $(p + q + 1)p > 0$
(C) $(p + q + 1)p = 0$ (D) Data Insufficient

18. If a, b, c are not all equal and $\alpha > 0$ such that $a^\alpha + b^\alpha + c^\alpha = 3$ and $f(\alpha) = 0$ then

- (A) $n = 2 - m$ (B) $n = 1 - m$
(C) $n = 3 - m$ (D) $m = n$

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

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